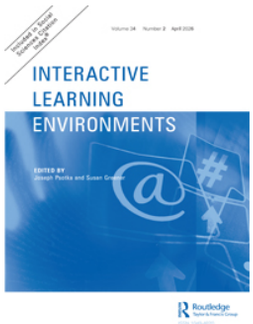

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G. Zwoliński, A. Dubiel & D. Kamińska

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Teaching empathy: a comparative analysis of real, virtual, and AI-driven methods

G. Zwoliński, A. Dubiel and D. Kamińska

Lodz University of Technology, Lodz, Poland

ABSTRACT
Empathy is a fundamental skill in universal design -- an approach that promotes inclusive, accessible environments for all people, regardless of ability, yet the most effective methods for teaching it remain an open question. This study examines three distinct approaches to empathy training, involving 64 students in a universal design course. Participants were divided into three groups, each following a different learning path: (1) role-playing experience using external devices such as wheelchairs, elderly suits, and pregnancy bellies, (2) a laboratory-based approach combining these tools with virtual reality, and (3) an AI-supported method, where the process is streamlined through AI-driven surveys. The article explores the full empathy training process, evaluating the strengths and limitations of each method. Additionally, it analyzes pre- and post-training empathy assessments completed by students, as well as their qualitative feedback on the effectiveness of each approach. The findings provide valuable insights into how role-playing, virtual, and AI-driven experiences contribute to empathy development.

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Empathy training; virtual reality; design education; large language models; educational technology

1. Introduction

Empathy plays a fundamental role in both everyday life and professional settings, influencing how individuals interact, make decisions, and design solutions for diverse populations (Decety & Cowell, 2014). It is frequently discussed as a desirable skill in various fields, from product development to healthcare (Batt-Rawden et al., 2013; Mattelmäki et al., 2014). However, the concept of empathy is often misunderstood and conflated with related terms such as compassion, self-transposition, pity, or identification with another person (Batson, 2009). These terms are often used interchangeably in literature and public discourse, raising the question: What exactly is empathy, and why is it so difficult to define? Historically, empathy has been a subject of philosophical discussion since ancient times (Stotland, 1978). The term itself was introduced over a century ago by Edward Titchener, who adapted the German word Einfühlung (Wispé, 1986). Despite its well-documented history, empathy still lacks a universally accepted definition, with researchers proposing various interpretations depending on the context (De Vignemont & Singer, 2006; Decety & Jackson, 2004). Some view it as a cognitive process, others as an emotional response, while many consider it a combination of both (De Vignemont & Singer, 2006; Decety & Jackson, 2004). Regardless of the specific definition, most researchers agree that empathy plays a crucial role in human relationships and social interactions. It is considered an essential personality trait that facilitates understanding, cooperation, and prosocial behavior (Kaźmierczak et al., 2007). Highly empathetic individuals are more likely to engage in helping behaviors, driven by what is known as "empathic motivation" (Gołab, 1976). This ability is closely linked to perspective-taking, emotional resonance, and altruism (Batson, 2014). Given its significance, empathy training has become a key focus in education and professional development, aiming to enhance emotional competencies that contribute to better interpersonal relationships, workplace collaboration, and prosocial behavior (Gerdes et al., 2010; Teding van Berkhout & Malouff, 2016). Developing empathy is a complex process that requires both cognitive and emotional engagement (Jones et al., 2015). Research suggests that effective empathy training involves experiential learning

CONTACT G. Zwoliński grzegorz.zwolinski@p.lodz.pl

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methods, which allow individuals to step into someone else's shoes and practice responses based on compassion and understanding (Kelm et al., 2014).

One of the most effective approaches to empathy education is role-playing and simulation-based training, where participants experience real-world challenges faced by others Eisenberg et al. (2002). These exercises help individuals develop a deeper emotional connection with different perspectives, reinforcing empathetic attitudes (Eisenberg et al., 2002). Another essential component is interpersonal communication training, which includes active listening and recognizing emotions in others Kafadar (2022). Research indicates that conversations rooted in active listening and mutual understanding foster empathy and build relationships based on trust and respect (Davis, 2018).

In recent years, virtual reality (VR) has emerged as a powerful tool for developing empathy, enabling participants to immerse themselves in experiences that replicate the challenges faced by others Ahn et al. (2013). Studies show that VR simulations can help individuals enhance their empathetic understanding by allowing them to experience physical limitations, social exclusion, or discrimination firsthand (Ahn et al., 2013). Unlike traditional role-playing, VR creates a highly immersive, controlled environment, making experiential learning more engaging and emotionally impactful (Herrera et al., 2018).

With the advancement of large language models (LLMs), researchers have begun exploring empathy training through AI-driven chat interfaces. Some propose utilizing chatbots to create "personas", virtual representations of various social groups, enabling users to gain insights into the challenges faced by different communities and fostering empathy and social awareness. This approach allows for deeper immersion into others' perspectives, enabling users to experience their daily realities and issues in a safe, controlled environment (Schuller et al., 2024).

This article examines the effectiveness of various experiential learning methods in empathy education, comparing real-world physical simulations, virtual reality experiences, and AI-driven analysis. To investigate their impact, we conducted a study with 64 participants, divided into three groups, each engaging in a different method. The study focused on how these approaches enhance perspective-taking and emotional engagement, providing insights into their potential for fostering empathetic understanding in educational and professional settings.

2. Methods for enhancing empathy

Empathy can be developed through various educational techniques, ranging from traditional approaches such as literature and role-playing to modern technological solutions like virtual reality (VR) and artificial intelligence (AI). Some methods require specialized training or advanced technology, while others rely on everyday experiences and cognitive engagement. This section reviews key methods used to teach empathy, highlighting their theoretical foundations and practical applications.

One established way to foster empathy is through reading fiction, which immerses individuals in characters' emotions and moral dilemmas. Nussbaum (1992) argues that following a character's journey shapes readers' thoughts and emotions beyond the reading experience. Oatley (1995) likens narrative immersion to a mental simulation of human actions, offering practice in perspective-taking. Neurocognitive studies confirm that reading activates brain regions involved in social interactions (Mar, 2011; Mar & Oatley, 2008), while Prasad (2016) highlights that narrative engagement requires both cognitive and emotional processing, enhancing empathic responses. Fiction also helps overcome biases by exposing readers to diverse perspectives, supporting theory of mind–the ability to understand others' beliefs, intentions, and emotions (Apperly, 2012).

Another method of teaching empathy is role-playing, which allows individuals to temporarily "step into someone else's shoes". A common example is adopting the persona of a well-known character, such as Superman, to explore different perspectives (Whitebread et al., 2012). Perspective-taking is a key component of empathic understanding, as it enables individuals to mentally and emotionally connect with others' experiences. Studies suggest that adopting another person's perspective reduces stereotypes and increases positive attitudes toward marginalized groups (Galinsky & Mussweiler, 2001). Kwon and Yawkey (2000) highlight the connection between role-playing, emotional intelligence, and psychoanalytic learning theories, emphasizing that simulated experiences can enhance self-awareness, emotional regulation, and social skills. This approach has also been integrated into digital role-playing games designed to foster empathy. For example, the game Real Lives was tested in high schools in the US, and researchers found that students

who played the game demonstrated greater global empathy and an increased interest in other cultures (Bachen et al., 2012).

Interactive digital experiences take role-playing a step further by introducing VR, which allows users to fully immerse themselves in another person's perspective. Studies show that VR can be a powerful tool for reducing bias and increasing empathy. For example, Peck et al. (2013) used VR to enable light-skinned participants to experience themselves as Black avatars, which significantly reduced negative implicit biases immediately after the experiment. Similarly, Banakou et al. (2016) found that this effect persisted for up to a week after exposure. Beyond racial bias reduction, VR has been used to simulate various social and physical challenges. De la Peña et al. (2010) placed participants in a VR experience of being a prisoner at Guantanamo Bay, where users reported emotional distress and a deeper understanding of human rights issues. The realism of VR simulations, achieved through multisensory engagement (visual, auditory, and tactile stimulation), enhances emotional impact and creates a sense of embodiment (Serino et al., 2015). The Proteus Effect, first described by Yee and Bailenson (2007), suggests that individuals' behavior and attitudes can change based on the appearance and characteristics of their virtual avatars. Research by Fox et al. (2013) and Peña et al. (2009) further confirms that virtual embodiment influences real-world attitudes, beliefs, and even moral decision-making.

Recent studies (Schuller et al., 2024) highlight the potential of LLMs in creating virtual personas that simulate diverse user experiences, providing learners with valuable insights into different social perspectives. Similarly, the course AI and Design Thinking offered by IDEO U demonstrates how AI can enhance empathy-building in the design process (IDEO U, 2025). By combining AI algorithms with design thinking methodologies, participants learn to generate virtual personas and simulated focus groups, enabling user experience prototyping in a controlled environment. This integration of AI and design thinking offers new opportunities for iterative, user-centered solution development before real-world implementation. Notably, platforms like the Interaction Design Foundation (Teleanu, 2025) demonstrate how LLMs can be utilized in the design process. This approach illustrates how AI can enhance empathy-building by enabling designers to create virtual personas and simulate user experiences, thereby gaining deeper insights into diverse user needs and emotions. Such integration of AI into user experience (UX) design fosters the development of user-centered solutions that are both innovative and empathetic.

While literature and role-playing remain valuable tools, technology offers new possibilities for immersive empathy training, enabling users to experience different perspectives in deeper and more interactive ways. Understanding the strengths and limitations of each method is essential for developing effective educational strategies that foster empathy in both personal and professional settings.

3. Materials and methods

3.1. Participants

All participants were first-semester master's students enrolled in the Human-Computer Interaction program at Lodz University of Technology, and the empathy training was conducted as part of a required course on Accessibility and Universal Design. The study took place during the first session of the course to eliminate potential biases from prior theoretical knowledge. Participation was voluntary, and students were informed that their involvement would not affect their final grades. All 64 students present agreed to participate in the study (approximately 75% men, 22% women, and 3% not specified; mean age = 20 years, SD = 1, median age = 20 years).

The research program received approval from the Ethics of Scientific Research from the Lodz University of Technology (approval no. #2/2021). All study procedures adhered to ethical standards, including obtaining full informed consent from all participants. Each participant was thoroughly informed about the purpose of the study, their right to withdraw at any time without providing a reason, and any other relevant details.

3.2. Procedure

Participants were randomly assigned to one of three empathy training modalities: real-world role-play, virtual reality (VR), or AI-driven dialogue. Each group followed only one method throughout the study. Within each

group, all students engaged with three predefined scenarios involving physical limitations–wheelchair use, aging-related decline, and late-stage pregnancy–using the assigned method. For example, participants in the AI group explored all three scenarios via persona-based dialogue with a large language model, while those in the VR and real-world groups experienced them through immersive simulations.

Although students did not rotate between methods, the study concluded with a two-stage reflective discussion. First, participants engaged in guided conversations within their small teams (3–4 students), supported by the instructor, to discuss their experiences and challenges in depth. This was followed by a full-group comparative session conducted in a plenary format, where all students from the three modality groups came together to exchange perspectives. The collective debriefing enabled cross-method insights and allowed students to contextualize their individual experiences within the broader spectrum of empathy training techniques.

All students completed pre-tests to measure baseline empathy levels, enabling comparison before and after the intervention. Each modality group, guided by an instructor, was further divided into teams of 3–4 to facilitate collaborative learning and discussion. This structure allowed students to explore individual limitations from multiple perspectives, reflect critically on their experiences, and share insights within their teams, fostering deeper empathy and awareness of accessibility challenges.

No fixed time limits were imposed on any of the activities. Participants proceeded at their own pace, and the durations varied between groups depending on the complexity and logistics of the modality. In particular, the real-world group required more time for transportation, preparation, and task execution in the store environment. This variability was intentional and preserved to reflect authentic user experience conditions.

3.3. Developing empathy through role-playing

One of the groups participated in a real-world shopping exercise designed to simulate physical limitations commonly experienced by different populations. Each student was assigned one of three conditions: using a wheelchair, wearing an aging suit, or donning a pregnancy simulator. This experiential approach aimed to provide firsthand insight into the challenges associated with mobility restrictions and their impact on routine activities such as grocery shopping. To ensure comparability with the virtual environment used by the second group, a standardized shopping list was developed, mirroring the obstacles present in the VR-based scenario. The list included items positioned at varying heights–on high shelves, at waist level, and near the floor–requiring participants to navigate the store while confronting mobility constraints and assessing their ability to perform seemingly simple tasks. Prior to entering the store, participants received detailed instructions regarding safety protocols, task objectives, and key observational aspects, including accessibility barriers, physical strain, and social interactions with fellow shoppers and store employees. Working in small subgroups, they were required to locate and select products, navigate the aisles, and complete the checkout process as they would in a typical shopping experience. Upon completing the exercise, participants filled out a post-test measuring empathy levels and documented their experiences in a written report. They recorded specific challenges encountered, moments of insight, and reflections on how physical limitations influence autonomy, mobility, and social engagement in public spaces. In the following session, students engaged in a structured discussion with the instructor, analyzing their findings and reflecting on the broader implications of accessibility and inclusion (Figure 1).

3.4. Developing empathy through virtual reality

The second group undertook the same shopping challenge in a virtual environment using Meta Quest 2 headsets and motion controllers, where participants simulated physical limitations also employing an elderly suit, pregnancy belly simulator, and wheelchair to deepen immersion. The session was conducted in a designated 10x5 meter area, ensuring a safe and controlled environment. The custom-built application (read more in our previous paper Kaminska et al. (2023), developed in Unity, allowed participants to navigate a virtual self-service store, with teleportation points facilitating movement between aisles. The selection of items for the virtual shopping task was carefully matched to the real-world store scenario, ensuring that products were placed at similar heights and posed comparable challenges and limitations. Before starting, participants followed similar procedures to those of the real-world store group, receiving comparable

Figure 1. Students participating in a role-playing session, from left to right: using a wheelchair, wearing a geriatric suit, and equipped with a pregnancy simulator.

instructions and task guidelines to ensure consistency between both experiences. Throughout the simulation, participants faced realistic challenges similar to those in the real-world store group, and upon completing the VR shopping task, they also completed a post-test assessing empathy levels and documented their reflections in a written report, followed by discussions with the instructor to explore key challenges and insights gained (Figure 2).

3.5. Developing empathy via AI-driven dialogue

The third group undertook a theoretical and analytical approach to empathy training, utilizing large language models (LLMs), specifically ChatGPT 4.0, to systematically explore and analyze the accessibility

Figure 2. Students participating in a VR session, from left to right: using a wheelchair, wearing a geriatric suit, and equipped with a pregnancy simulator.

challenges faced by individuals with physical limitations in a typical shopping environment. As with the other groups, participants began with detailed instructions highlighting key learning objectives, which included the identification of common accessibility barriers experienced by wheelchair users, elderly individuals, and pregnant shoppers.

In the initial stage, students collaborated with the AI to develop detailed user personas representing various physical impairments, formulating their inputs independently without the use of predefined prompts. These personas were meticulously crafted, outlining not only basic demographic data but also deeper cognitive and emotional aspects such as individual fears, needs, preferences, and typical responses to encountered barriers. Through iterative dialogue with the AI, students enhanced the realism and complexity of these personas, allowing them to better grasp the nuanced experiences of individuals facing mobility and accessibility challenges.

Participants engaged with the LLM via a standard web interface, working in groups without fixed time limits, with the objective of gaining as much insight as possible. No fixed templates were provided, and students were encouraged to formulate their own prompts. For ethical consistency, the instructions emphasized the use of respectful and non-stereotypical language, the avoidance of personal data, and a focus on actionable, person-centered improvements. The original prompts were written in the participants' native language; however, the examples shown in the Appendix are English translations prepared for clarity while preserving the original structure and intent. Within their groups, students collaboratively explored scenario narratives generated by ChatGPT, simulating typical accessibility challenges such as navigating crowded aisles, reaching products on high shelves, interacting with store staff, and handling unexpected obstacles. These activities were conducted through structured textual dialogue and analytical discussion with the AI system. In addition to interacting with the model, students discussed the AI-generated responses within their teams, critically reflecting on the realism, relevance, and potential design implications of each scenario. Through these explorations, they evaluated how physical impairments affected decision-making processes, movement efficiency, autonomy, and overall satisfaction with the shopping experience. No physical or in-person role-play was involved in this condition.

Following their research, students completed a post-test on empathy and compiled their findings in reports (accompanied by prompt transcripts), concluding with discussions on accessibility improvements and inclusive design, similar to the processes followed by the previous groups.

3.6. Assessing empathy development across groups

We employed two measurement tools to assess the empathy levels of all participant groups. The first tool was the Davis Interpersonal Reactivity Index (IRI) (Davis, 1980), which measures multiple dimensions of empathy and served to assess participants' baseline empathy levels before the exercises and their progress afterward. IRI is a 28-item questionnaire rated on a 5-point Likert scale, consisting of four subscales, each with seven items: Perspective Taking PT (adopting others' psychological viewpoints), Fantasy Scale FS (imagining oneself in the feelings and actions of fictional characters), Empathic Concern EC (sympathy for others in distress), and Personal Distress PD (anxiety in tense interpersonal situations). The second tool was a custom-developed questionnaire designed specifically for this research, focusing on empathy toward the created personas, including individuals using wheelchairs, the elderly, and pregnant women (see the Appendix 2). This dual approach provided a comprehensive analysis of participants' empathetic responses across different social groups.

Following the completion of the exercises, participants engaged in structured group reflections conducted within their small teams. These discussions, facilitated by the instructor, were audio-recorded with participants' consent, transcribed verbatim, and subjected to thematic analysis. The aim was to capture students' subjective perspectives on the learning experience across different modalities, including emotional engagement, perceived realism, and suggestions for improvement. Subsequently, a joint plenary discussion was held with all participants to enable comparative reflection across the three empathy training approaches. This full-group exchange fostered a broader understanding of empathy-building strategies and allowed participants to articulate key takeaways from their respective experiences.

4. Results

All pre- and post-intervention data series were first assessed with the Shapiro–Wilk test, yielding *p*-values that provided no grounds to reject the null hypothesis of normality. Subsequently, paired *t*-tests–and, for robustness, Wilcoxon signed-rank tests–confirmed statistically significant increases for every analyzed indicator. Full numerical results for the Shapiro–Wilk, paired *t*-, and Wilcoxon tests are reported in Appendix 3. As shown in Figure 3, all three groups demonstrated significant increases in empathy scores across the PT, FS, EC, and PD categories following their respective interventions. The role-playing (RP) group exhibited the greatest improvement, with IRI overall increase of 13.2% (especially in the PT and PD categories, achieving increases of 13.5% and 13.6%, respectively), indicating that physically simulating constraints in a real environment can substantially enhance empathic responses. The VR group also achieved significant gains, particularly in the PT and FS categories, although its IRI increase (11.6%) was slightly lower than that of the RP group. The AI group, despite showing the smallest IRI increase (8.8%), still demonstrated a notable positive change in empathy measures, suggesting that engaging with AI-driven scenarios can foster empathic awareness, albeit to a lesser extent than more immersive methods.

Results from the custom-developed questionnaire focusing on specific physical limitations (PB – pregnancy belly, GS – elderly suit, and W – wheelchair users) further support these findings. The RP group consistently showed the highest empathy gains across these targeted categories (16.2%, 14.1%, and 14.8%, respectively), highlighting the effectiveness of the role-playing experience. The VR group followed closely, particularly in the PB and W categories, while the AI group exhibited the most modest improvements. These results reinforce the value of immersive experiences, both physical and virtual, in fostering empathy toward specific physical challenges.

Table 1 summarizes the statistical analyses of the pre-to-post increases in each empathy dimension, comparing those changes across the three groups. The primary goal of these analyses was to determine whether the choice of empathization method would have a measurable effect on the four IRI subscales (PT, FS, EC, and PD) as well as on the three state-specific empathy measures (PB, GS, and W).

Normality was assessed using Shapiro–Wilk tests, all yielding non-significant *p*-values, ensuring the data met parametric assumptions. Levene's tests confirmed homogeneity of variances across groups. A one-way ANOVA indicated no significant differences for PT, FS, and EC, though PT and EC showed trends ($p = 0.0563$ and $p = 0.087$). Significant differences were found in PD ($F(2, 61) = 7.595$, $p = 0.0011$), with post-hoc Tukey tests highlighting significantly higher RP scores compared to AI and lower VR scores compared to RP.

While there is a trend for the RP and VR groups to score higher than the AI group in the PT and EC categories, these differences did not reach statistical significance at the $\alpha = 0.05$ level. In contrast, the PD category showed statistically significant differences, with the RP group exhibiting markedly higher levels of personal distress compared to the AI group, as well as significant differences between RP and VR. These

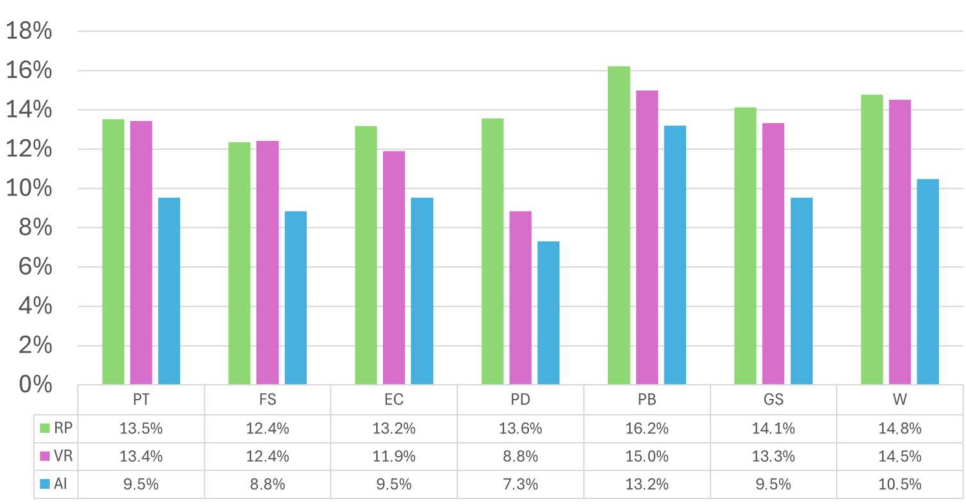

Figure 3. Increase in empathy level based on two different questionnaires: Davis Interpersonal Reactivity Index (PT, FS, EC, PD) and custom-developed questionnaire focusing on empathy toward the created personas (PB, GS, W).

Table 1. Statistical analysis of pre–to–post changes in empathy dimensions across the RP, VR, and AI groups (two questionnaires).

Type	Normality (p)	Levene (p)	ANOVA (F, p)	Post-hoc (p)
PT	0.129–0.739	0.911	3.017, 0.056	RP vs AI: 0.107, VR vs AI: 0.080
FS	0.056–0.168	0.181	1.670, 0.197	No significant differences
EC	0.305–0.455	0.549	2.542, 0.087	RP vs AI: 0.070
PD	0.158–0.414	0.745	7.595, 0.001	RP vs AI: 0.002, VR vs RP: 0.013
PB	0.117–0.454	0.322	0.706, 0.498	No significant differences
GS	0.068–0.135	0.154	1.439, 0.245	No significant differences
W	0.082–0.102	0.418	1.022, 0.366	No significant differences

Table 2. Illustrative student reflections by training modality and theme.

	Theme	Representative Quotes
VR	Emotional impact	"Wearing the pregnancy belly in VR made me realize how even simple movements like bending or turning become complicated." "It was a strange feeling – my body wasn't really old, but my movements were limited. I felt frustrated like someone much older." "The simulation helped me understand how quickly everyday independence can be lost."
VR	Realism	"The aging suit made me feel how limited mobility affects everything – I suddenly understood what joint stiffness really means." "I couldn't keep my balance well and it made me think about how dangerous stairs or uneven sidewalks might be."
VR	Challenges	"The headset was heavy and made me dizzy after a while, which distracted me from the scenario." "It was hard to focus when the view was slightly blurry, and I felt a bit nauseous." "I wish there were more senses involved – like touch or even smell – it still felt like watching, not living it."
RP	Emotional impact	"Pushing the wheelchair through narrow aisles was exhausting – I didn't expect it to be so hard just to move around." "After walking with the aging suit for 20 minutes, I was tired and annoyed. It gave me insight into how frustrating life can be." "I felt vulnerable doing simple things. That emotion stuck with me the most."
RP	Realism	"People offered help when they saw me in a wheelchair. That surprised me and made the experience feel more real." "Someone held the door open for me, and it made me realize how much people with disabilities rely on others." "It felt like real life – there were noises, smells, people. Not like in a lab or classroom."
RP	Challenges	"It took a lot of time and energy – more than I expected. It was tiring and not very practical to repeat." "I got physically exhausted, and that made it hard to stay focused on the learning part." "Logistically it's difficult – we can't go out and repeat that kind of thing easily."
AI	Analytical insight	"Creating a persona forced me to think more deeply about their emotional state and needs." "I started asking questions like 'what does she feel when she can't reach the shelf?' – that was new for me." "It helped me reflect on situations I wouldn't have thought about otherwise."
AI	Challenges	"I understood the challenges intellectually, but without feeling them physically, it wasn't the same." "It felt more like a theoretical exercise – helpful, but distant." "I think it would work better if we could combine the AI with something sensory or visual."

findings suggest that the RP method may evoke stronger emotional responses (as measured by PD) compared to the other methods, whereas differences in the other dimensions of empathy are less pronounced.

However, when examining the results from the custom-developed questionnaire focused on specific physical conditions (PB for pregnancy, GS for elderly limitations, and W for wheelchair users), no significant differences were observed between the groups. Although all groups demonstrated an increase in empathy after the interventions, the statistical analyses (ANOVA and Tukey HSD tests) revealed no measurable advantage of one intervention method over another in these specific areas.

The average task durations in the three groups were 125 $\pm$ 4.38 min, 87 $\pm$ 3.90 min, and 42 $\pm$ 2.90 min respectively, and with such a small sample size (6 per group), there is no clear evidence to reject the assumption of normality.

Table 2 presents selected student reflections gathered during post-activity group discussions. The quotes are grouped by training modality and organized according to key thematic categories identified through thematic analysis. These categories include emotional impact, perceived realism, encountered challenges, and analytical insight. The excerpts illustrate the range of participants' subjective experiences and perceptions, highlighting both the strengths and limitations of each approach to empathy training.

5. Discussion

5.1. Effectiveness of empathy training methods

The results of both the IRI and the custom-designed questionnaires indicate that all three approaches–real-world role-play, virtual reality (VR), and AI-based analysis–had a measurable impact on students' empathy

levels. The role-play group showed the greatest overall improvement, particularly in the *Perspective Taking* (PT) and *Personal Distress* (PD) subscales. Importantly, the change in PD was an *increase* rather than a decrease, suggesting heightened sensitivity to others' discomfort rather than reduced egocentric stress. This suggests that physically simulating limitations in real-world settings can evoke stronger cognitive and emotional engagement because embodied interactions provide multisensory feedback and activate mechanisms posited by embodied-cognition theory, thereby deepening self–other overlap.

The VR group demonstrated moderate gains, especially in the PT, *Fantasy* (FS), and *Empathic Concern* (EC) dimensions, aligning with existing research on virtual embodiment and affective resonance; these effects are thought to arise from the "body-ownership" illusion and the first-person perspective afforded by immersive VR, which together enhance affective sharing. Although the AI group exhibited the smallest increase in empathy scores, the improvement remained statistically significant (see Appendix 3 for full statistics). This finding highlights the potential of analytical, dialogue-based approaches to stimulate cognitive perspective-taking even in the absence of physical immersion.

These results are consistent with prior studies suggesting that immersive and embodied experiences, such as VR and role-play, tend to elicit stronger empathic responses than abstract or purely analytical methods (Ahn et al., 2013; Banakou et al., 2016; Herrera et al., 2018). For example, Herrera et al. (2018) found that VR can promote long-term increases in empathy, while Banakou et al. (2016) demonstrated that virtual embodiment can reduce prejudice. Kelm et al. (2014) emphasized that empathy training is most effective when emotional engagement is paired with reflective analysis. Our findings align with this perspective: although the AI-based condition lacked physical immersion, it fostered meaningful cognitive reflection. Notably, EC scores rose significantly in both the role-play and VR conditions–likely because direct or simulated face-to-face interaction facilitates emotional contagion and compassionate concern–reinforcing the importance of affective components in empathy training. Overall, the results support the view that multimodal interventions–especially those integrating both emotional and cognitive components–are particularly effective in promoting and teaching empathy.

5.2. Logistical considerations

From a logistical standpoint, the real-world shopping experience presented the most complex challenge, primarily due to the necessity of transporting the required equipment to a nearby shopping center with a large self-service store. Consequently, each subgroup navigating the physical store spent the most time in this setting–an average of approximately 125 minutes, depending on whether the group comprised three or four students. This extended duration allowed for a thorough exploration of mobility and accessibility barriers but demanded significant coordination and setup.

By contrast, the VR session was conducted in a dedicated lab, where all the necessary equipment was readily available. Although preparing the VR headsets and motion controllers and providing detailed instructions took a substantial amount of time, the entire session lasted about 87 minutes on average–noticeably shorter than the real-world experience. The controlled environment and immediate access to technology streamlined the practical aspects of the exercise, even though participants required thorough briefings on using the VR system effectively and continuous assistance from the instructor due to occasional application freezes or accidental exits from the experience.

The AI-supported analysis proved to be the most logistically efficient approach. Both the introductory briefing and the subsequent analytical exercise were completed relatively quickly, averaging around 42 minutes. With no need to travel offsite or maneuver specialized equipment, students could focus their efforts on exploring AI-generated insights, while minimizing the workload and time commitment required from the instructor. This expedited process not only highlighted the analytical dimension of empathy-building but also underscored the potential advantages of AI in facilitating swift yet meaningful reflections on accessibility and inclusive design.

5.3. Student feedback from comparative discussion

During the comparative discussion, students provided detailed reflections on their experiences with each method of empathy training, offering critical insights into the strengths, limitations, and potential

improvements for each approach. VR simulations provided a structured and cost-effective platform for replicating physical limitations through scenarios such as pregnancy, aging, and wheelchair use. Students highlighted the realism of physical constraints, such as limited mobility, joint stiffness, and balance challenges. Some noted that the pregnancy simulation made them more aware of how routine tasks become complex, while others expressed that the aging suit helped them empathize with the gradual decline in physical abilities.

Challenges included the lack of immersive multi-sensory feedback within the virtual environment itself–beyond sight and sound–even though participants wore physical simulation gear. Additional limitations included technical issues such as headset discomfort and occasional dizziness. Despite these limitations, students expressed strong interest in the VR-based approach and indicated a willingness to engage with it more frequently. They believed that regular exposure to such simulations could significantly enhance their empathy and deepen their understanding of accessibility-related challenges.

Real-world experiences were praised for their authenticity, offering full sensory involvement, including tactile interactions, spatial awareness, and real-time social interactions. Students noted that pushing a wheelchair through narrow aisles or experiencing the physical strain of bending repeatedly gave them a clearer understanding of the difficulties faced by individuals with physical impairments. Many also highlighted the kindness and willingness of people to offer assistance, particularly when using a wheelchair, which added a positive and encouraging aspect to the experience. However, some mentioned that the time-intensive nature of real-world tasks and the physical effort required were challenging, particularly when compared to digital methods.

The LLM-based method provided students with an analytical approach to empathy training through user personas. Students found this method particularly useful for understanding the cognitive and emotional challenges faced by individuals with disabilities. Some reflected that creating personas made them more mindful of accessibility in design, while others felt that without physical immersion, their connection to these challenges was limited. Suggestions included integrating AI with VR for a holistic learning experience

Overall, the discussion revealed that each method contributed distinctively to empathy development: VR offered immersive yet flexible simulations, real-world experiences provided authentic and sensory-rich interactions, and LLM-based analysis delivered scalable insights. Students concluded that combining elements of all three methods could bridge the gap between emotional immersion, sensory engagement, and analytical exploration, offering a comprehensive framework for future empathy training programs. They also recommended that future research focus on hybrid models that leverage VR's repeatable scenarios, real-world's tangible experiences, and LLM's adaptive analysis, with improvements in VR technology and AI-driven feedback to enhance empathy education effectiveness.

5.4. Limitations

While our findings offer promising insights, several limitations should be acknowledged. First, the study was conducted with a relatively small and homogeneous sample–64 students from a single institution, all enrolled in the same course. This convenience sample, combined with a notable gender imbalance, limits the generalizability of our results across age groups, educational backgrounds, and cultural contexts.

Second, although instructional goals were aligned across the three groups, the actual duration of each activity varied considerably: approximately 125 minutes for the real-world task, 87 minutes for VR, and 42 minutes for the AI-based task. While no time limits were imposed, this natural variation introduces a confounding factor, making it difficult to attribute outcome differences solely to instructional modality rather than exposure time or cognitive effort.

Third, the practical implementation of each method posed distinct challenges. The real-world scenario required complex logistics, including equipment transport, coordination in public spaces, and longer sessions, which may be difficult to replicate in time-constrained educational settings. Similarly, the VR sessions, although conducted in a controlled lab, required significant setup time (calibration, hardware preparation) and continuous technical support from instructors. Some participants also reported physical discomfort, such as dizziness and headset fatigue, as well as limited sensory immersion due to the dominance of visual and auditory channels. In contrast, the AI-based approach was time-efficient and cognitively engaging,

but lacked emotional or physical immersion, which some students perceived as limiting for empathy development.

Fourth, all methods relied on active instructor involvement, particularly in the VR and AI groups, where technical issues or lack of prompting guidance could significantly affect the learning process. This limits the potential for independent deployment without adequately trained staff.

Fifth, although the scenarios were consistent across modalities–focused on pregnancy, aging, and wheelchair use in a grocery shopping context–the study did not include other important domains such as sensory, cognitive, or invisible disabilities, nor other real-world environments like workplaces or public transport. This limits the ecological validity and scope of the findings.

In terms of measurement, while we used the validated Interpersonal Reactivity Index (IRI), we also developed a custom questionnaire tailored to each group's activity. Although it captured context-specific reflections, it has not yet undergone psychometric validation, which raises concerns about reliability and sensitivity to change.

Finally, although all groups showed increased empathy scores, statistically significant improvements emerged primarily in the *Personal Distress* subscale. Other subscales demonstrated only non-significant trends. Given the sample size and multiple comparisons, the possibility of Type II error cannot be ruled out. Furthermore, the unstructured nature of real-world social interactions–such as spontaneous help from bystanders–may have amplified emotional responses in the RP group in ways that were not fully matched in the more controlled VR or AI environments.

5.5. Future directions

To build on these findings and enhance their applicability, we plan a multisite randomized controlled study with time-matched conditions and standardized instructor interaction across the three modalities. Outcome measures will include the full IRI scale, state-based empathy assessments, behavioral indicators of prosocial action (e.g. performance in inclusive design tasks), and longitudinal follow-up (1–6 months) to assess knowledge retention and transfer to real-world behavior.

In parallel, we aim to develop a hybrid learning pipeline that combines cognitive, immersive, and experiential layers. Learners will begin by generating personas and scenario variants using AI, continue by interacting with parameterized VR environments derived from those personas, and conclude with micro-immersions in real-world settings supported by structured debriefings. We hypothesize that combining perspective-taking with embodied experience will lead to deeper and more durable empathy development.

At the methodological level, future work will include standardization of the AI component via prompt libraries, anti-stereotyping guardrails, and reproducible workflows (e.g. persona templates, rubric-based quality checks). We also plan to psychometrically validate the custom empathy questionnaire (via EFA/CFA, reliability and test–retest analysis) to enable its use beyond the grocery shopping context.

Our scenario coverage will be expanded to include sensory, cognitive, and intersecting disabilities, and new domains such as public transport, healthcare access, and digital services. To improve generalizability, we will recruit more diverse participants, including non-students and individuals from multiple cultural backgrounds. Planned improvements in the VR component include enhanced fidelity and accessibility (e.g.multisensory cues, improved locomotion, mitigation of cybersickness, and safeguards against session interruption).

Finally, we plan to reflect on the resource demands of each modality by analyzing observed time footprints (approximately 125/87/42 minutes) and estimating staff involvement, equipment needs, and physical space requirements. We aim to make selected teaching materials and anonymized example prompts available to educators interested in adopting similar approaches. We also hope to engage with students and accessibility advocates to refine scenario content and ensure its relevance across diverse user groups.

Together, these efforts aim to produce robust, scalable, and cost-effective empathy training frameworks that can be integrated into a wide range of educational and professional contexts. Our goal is to deliver time- and cost-matched effect estimates across modalities, validate transferable assessment tools, and demonstrate lasting behavioral change that supports inclusive design practices. Ultimately, we hope to provide educators, designers, and institutions with actionable models for empathy training that are grounded in both scientific evidence and practical impact.

5.6. Implications for practice

This study highlights several practical ways in which educators, designers, and policymakers can apply empathy training methods to real-world settings.

First, educators can incorporate empathy-building exercises into design, healthcare, or engineering curricula without requiring complex equipment. For example, the AI-supported method offers a scalable, low-cost approach that can be implemented in classrooms using only web access and brief facilitator guidance. It allows students to explore user perspectives, co-create inclusive design ideas, and reflect critically on social barriers.

Second, the findings suggest that virtual reality (VR) offers a balanced trade-off between realism and feasibility. Educational institutions with access to VR equipment can use it to create emotionally engaging scenarios, particularly useful in inclusive design education, medical training, or social work programs. Reusable VR scenarios–such as a virtual grocery store–enable repeatable and standardized learning experiences.

Third, policymakers and training providers in professional settings can use the results to justify broader adoption of hybrid empathy training models. For example, combining AI-driven persona work with targeted VR simulations or short real-world immersions could improve empathy and inclusion awareness among employees in public services, healthcare, and retail.

Example: In a technical university setting, engineering students could use large language models to design user personas representing individuals with physical disabilities. Based on those personas, students would then prototype assistive devices or modify public space layouts. These proposals could later be tested in a VR environment simulating real-world constraints such as narrow aisles or high shelves. This approach not only deepens students' empathy but also links directly to practical problem-solving and user-centered design skills.

Across all contexts, a key takeaway is that empathy can be taught using different methods depending on available resources, time, and learning goals. While real-world simulations yield the strongest emotional impact, even AI-based methods–if designed thoughtfully–can promote meaningful reflection on user needs and equity.

To support implementation, we encourage instructors and organizational leaders to:

- Select methods appropriate for their learners' needs and logistical constraints.
- Use empathy-building not only to raise awareness but also to inspire actionable design improvements.
- Create space for guided discussion and reflection after each activity.
- Involve individuals with lived experience (e.g. disability advocates) in designing and evaluating training materials.

By aligning empathy training with concrete application goals–such as inclusive product development or accessible public services–stakeholders can ensure that learning translates into real-world impact.

6. Conclusion

This study demonstrates that all three methods of empathy training – real-world role-playing, virtual reality simulations, and AI-driven analysis – effectively enhance empathy among participants. The findings highlight that while immersive, hands-on experiences yield the highest empathy gains, VR simulations provide a balanced alternative due to their flexibility and repeatability. AI-based methods, although less immersive, offer scalable and cost-effective opportunities for fostering empathy, particularly in settings with limited resources.

The comparative analysis underscores the importance of selecting an empathy training method based on specific educational goals, available resources, and time constraints. Furthermore, there remains significant untapped potential in integrating these methods. For example, future studies could explore hybrid approaches where participants first engage in AI-driven dialogues to deepen their cognitive and emotional understanding of disabilities, followed by immersive VR simulations informed by these dialogues. A hypothetical scenario might involve participants collaboratively developing detailed emotional and cognitive profiles of virtual personas using AI, which would then directly shape personalized VR experiences, such as navigating a shopping environment tailored to these personas' specific needs and challenges. Such integrated frameworks would enhance the realism and impact of empathy training, enabling participants to

combine analytical reflection with emotional immersion more effectively. Future research should evaluate these hybrid models and assess their long-term impact on participants' empathetic behaviors in real-world contexts. This study contributes valuable insights to the growing body of research on empathy training, offering practical guidance for educators, designers, and policymakers seeking to develop empathy through innovative pedagogical tools.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Notes on contributors

Grzegorz Zwoliński received his M.Sc. and Electrical Engineer degrees from the Institute of Mechatronics and Information Systems at Lodz University of Technology (TUL) in 1992, and earned his Ph.D. degree from the same institution in 1999. Since 1993, he has served as an academic teacher at TUL, and since 1995 he has also been affiliated with the University of Computer Sciences and Skills in Łódź.

His research interests combine technical and creative disciplines, spanning electrical machines, computer graphics, graphics programming, and the development of interactive systems. He is the author and lead developer of multiple IT management platforms, computational tools for analyzing electrical machines, and real-time 3D graphics engines.

In recent years, his focus has shifted toward immersive technologies. He is currently involved in several international projects exploring the application of virtual and augmented reality in education, universal design, and social impact contexts. As a researcher and practitioner, he bridges engineering precision with user-centered design, contributing to interdisciplinary advancements in interactive systems and experiential learning.

Agnieszka Dubiel graduated with a degree in Human-Computer Interaction from the Lodz University of Technology. She is currently continuing her education as part of her PhD studies in the specialization of Information and Communication Technology. Member of the Voxel Research Lab operating at the Institute of Mechatronics and Information Systems of the Lodz University of Technology. She is focusing her research activities on combining modern VR and AR technologies with user responses collected as biomedical signals. She has been involved in many international interdisciplinary projects for several years. Additionally, she has received several national and European research grants to pursue her scientific passions. She was the main organizer of the FG24 challenge and workshop. She is passionate about biomedical signal analysis, practical applications of usability testing, customizing technology for user preferences and needs, and using machine learning to automate data collection. She is constantly looking for new challenges, pursuing scientific dreams and finding opportunities for continuous self-development.

Dorota Kamińska graduated in Automatic Control and Robotics and completed postgraduate studies in Biomedical image processing and analysis at Lodz University of Technology. She received her PhD degree from Faculty of Electrical, Electronic, Computer and Control Engineering at Lodz University of Technology in 2014. She gained experience during the TOP 500 Innovators program at Haas School of Business, University of California in Berkeley. An educator and researcher at Institute of Mechatronics and Information Systems, a founder and a leader of the Voxel Research Lab, currently focusing her academic pursuits on cutting-edge VR and AR applications. She is an Associate Editor and Guest Lead Editor of several journals and Special Issues such as Electronics, Entropy, Information. She has received several national and European Research grants and has been involved in many international industrial projects. She has published over 50 scientific works and has been organizing challenges and workshops in ECML19 and FG20. She is passionate about biomedical signal analysis and practical applications of machine and deep learning. As a participant of many interdisciplinary and international projects, she is constantly looking for new challenges and possibilities of self-development.

ORCID

G. Zwoliński http://orcid.org/0000-0002-6100-2654
A. Dubiel http://orcid.org/0000-0001-8758-7593
D. Kamińska http://orcid.org/0000-0002-3416-5554

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Appendices

Appendix 1. Exemplar AI prompt sequence (english translation)

Student (1): Good morning. Could you adopt the role of a person who uses a wheelchair so that I can better understand your needs?

Persona (2): Good morning. Certainly. I will assume the role of Mr Adam Nowicki, a 38-year-old active-wheelchair user following a T10 spinal-cord injury ten years ago. I will gladly answer your questions to help you build empathy.

Student (3): Thank you. Could you provide a short project-profile: the key facts, priorities, and constraints of your everyday life?

Persona (4):

- **Family status:** married, one child aged 6.
- **Occupation:** data analyst, hybrid work model.
- **Mobility:** manual wheelchair; I transfer independently to my car and drive with hand controls.
- **Shopping priorities:** speed, independence, safe access to every section of the store.
- **Constraints:** cannot reach above $\approx$ 140 cm; thresholds $>$ 2 cm are barriers; aisles $<$ 90 cm are problematic.
- **Preferred features:** wide aisles, lowered self-checkout counters, in-store map in the mobile app.

Student (5): What does a typical supermarket visit look like for you and what emotions accompany it?

Persona (6): I usually shop on Tuesday evenings when the store is less busy. At the entrance I look for *trolley-basket* carts with extended handles. If none are available, I feel anxious about being late to pick up my child from after-school care. The process is satisfying when I can handle everything myself but frustrating when, for instance, promotions are placed only on top shelves.

Student (7): Which stages of shopping do you find the most problematic?

Persona (8):

- **Reaching items on high shelves** – shoulder-level reaching causes pain; I often need to ask other customers for assistance.
- **Self-checkout zone** – terminals are sometimes mounted too high, and the scale platform will not fit a basket next to my chair.

Student (9): Regarding signage and in-store navigation–what works and what does not?

Persona (10): High-contrast wheelchair pictograms at lifts and accessible toilets are effective. Entrance maps, however, seldom show step-free alternative routes. Directional arrows are often on the floor, which I cannot always see from a seated position, especially when the store is crowded.

Student (11): Which store solutions or features do you particularly value?

Persona (12):

- Pull-out shelves with products no higher than 130 cm.
- The ability to scan items with a mobile app directly in the aisle.
- Folding mirrors in the dairy section so I can read expiration dates on items placed at the back.

Student (13): And what discourages you most from a store?

Persona (14): Tight turns between promotional islands, a missing ramp at a side entrance, and staff who insist on helping before I even ask. I want the option of help, not the assumption that I am automatically dependent.

Student (15): Do you use any digital tools that support your shopping?

Persona (16): Yes: the "Bring!" shopping-list app integrated with my family calendar, and Google Pay on my smartwatch so I don't juggle a wallet. Occasionally I reserve click-and-collect, but I prefer to inspect fresh goods myself.

Student (17): How do you balance your desire for independence with accepting staff assistance?

Persona (18): A proactive yet non-intrusive attitude is key. Ideally, an employee approaches, introduces themselves, and says, "Please let me know if you need any assistance." That gives me control, and it's easier to ask when I truly need help.

Student (19): Finally: what three recommendations would you give designers upgrading a supermarket for accessibility?

Persona (20):

- **Design from entry to exit** – continuous step-free routes at least 120 cm wide.
- **The two-reach rule** – every promotional item and staple category also available below 130 cm.
- **Technology that serves autonomy** – an app with a product locator and high-contrast mode, plus a fully extendable payment terminal at self-checkout.

Appendix 2

Table A1. Pre-test and post-test statements for empathy exercises related to pregnancy, elderly people, and wheelchair users. Authors of this table are Edyta Klimczak (Nofer Institute of Occupational Medicine, Lodz, Poland), Kaja Staszewska (Nofer Institute of Occupational Medicine, Lodz, Poland), and Dorota Merecz-Kot (University of Lodz, Lodz, Poland).

Exercise	Pre-test Statements	Post-test Statements
Pregnancy belly	I can put myself in the shoes of a pregnant woman. (✓)	I can put myself in the shoes of a pregnant woman. (✓)
	I do not understand why I should let a pregnant woman through in a queue. (X)	I do not understand why I should let a pregnant woman through in a queue. (X)
	I can predict what may be difficult for pregnant women. (✓)	I can predict what may be difficult for pregnant women. (X)
	When I see a pregnant woman I become more caring. (✓)	I better understand the problems of pregnant women. (✓)
	My perception of pregnant women has changed. (✓)	
	I care more about problems of pregnant women. (✓)	
Geriatric suit	Usually, I do not know how to deal with an elder person. (X)	I can predict what may be difficult for elder people. (✓)
	Elder people evoke negative emotions in me. (X)	My perception of elder people has changed. (✓)
	I can put myself in the shoes of an elder person. (✓)	I can put myself in the shoes of an elder person. (✓)
	I willingly help elder people. (✓)	I care more about elder people's problems. (✓)
	I do not understand why I should give place to an elder person. (X)	I do not understand why I should give place to an elder person. (X)
Wheelchair	I would spontaneously help a person who uses a wheelchair. (✓)	I can predict how a wheelchair user will feel in certain situations. (✓)
	People who use wheelchair arouse negative emotions in me. (X)	I can put myself in the shoes of a wheelchair user. (✓)
	I can predict how a wheelchair user will feel in certain situations. (✓)	People who use wheelchair arouse negative emotions in me. (X)
	It is difficult for me to assess whether I behave properly towards people who use a wheelchair. (X)	My perception of people using wheelchair has changed. (✓)
	I can put myself in the shoes of a wheelchair user. (✓)	I would spontaneously help a person who uses a wheelchair. (✓)

Appendix 3

Table A2. Comparison of statistical results (normality, paired t-test, and Wilcoxon test) across scales and methods (RP, VR, AI).

		RP		VR		AI	
Scale	Test	Before	After	Before	After	Before	After
PT	Shapiro-Wilk	0.7295	0.5840	0.4705	0.4590	0.7476	0.7000
	paired t-test	0.000001		0.000012		0.000007	
	Wilcoxon test	0.000001		0.000378		0.000253	
FS	Shapiro-Wilk	0.0745	0.0608	0.0745	0.0608	0.0853	0.0750
	paired t-test	0.000005		0.000078		0.000002	
	Wilcoxon test	0.000254		0.000725		0.000320	
EC	Shapiro-Wilk	0.0746	0.0609	0.0876	0.0349	0.0746	0.0694
	paired t-test	0.000005		0.000006		0.000001	
	Wilcoxon test	0.000232		0.000542		0.000194	
PD	Shapiro-Wilk	0.0994	0.0876	0.0180	0.0137	0.3189	0.3029
	paired t-test	0.000001		¡0.000001		0.000003	
	Wilcoxon test	0.000068		0.000171		0.000246	
PB	Shapiro-Wilk	0.4541	0.1778	0.0672	0.2068	0.9346	0.1679
	paired t-test	0.000001		0.000001		0.000001	
	Wilcoxon test	0.000001		0.000001		0.000031	
GS	Shapiro-Wilk	0.2857	0.5674	0.1477	0.6266	0.4642	0.2506
	paired t-test	0.000001		0.000001		0.000001	
	Wilcoxon test	0.000001		0.000123		0.000002	
W	Shapiro-Wilk	0.0995	0.0776	0.0088	0.0091	0.2004	0.2183
	paired t-test	0.000001		0.000001		0.000001	
	Wilcoxon test	0.000198		0.000298		0.000221	